African Ungulate Life Tables
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AFRICAN UNGULATE LIFE TABLES[1]

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Abstract. The survival curves of four species of African ungulate—impala, buffalo, zebra, and warthog—derived from skulls found in the Akagera National Park, Rwanda, are compared. Three of the species are shown to have the typical logarithmic negatively-skewed adult curve shown by undisturbed large mammal populations, but warthogs apparently have a heavy age-constant number of deaths to middle-age. Attention is drawn to the limitations of time-specific data in the study of population dynamics.

Between July, 1968, and September, 1969, a collection of skulls of large ungulates that had died natural deaths was made in the Akagera National Park, Rwanda. Data from these skulls were used to construct time-specific life tables. The limitations of such life tables in population studies are discussed.

Four species were examined in detail: impala *Aepyceros melampus* Lichtenstein, buffalo *Syncerus caffer* Sparrman, zebra *Equus burchelli boehmi* Matschie, and warthog *Phacochoerus aethiopicus* Pallas. Comparisons were made with published data for other species.

An ecological outline of the area has been given by Troupin (1966) and Spinage and Guinness (1972).

METHODS

Skulls were recovered during travels through the area, and searches were made by vehicle after the annual burning, when visibility was extremely good. Soils were acid, causing rapid deterioration of the skull, and it was estimated that none was collected older than 2 years post mortem.

Ages at death were determined by differing methods. Juvenile impala ages were determined according to the data of Child (1964), adult ages according to Spinage (1971). Buffalo ages were determined from measurement of the height of the first maxillary molar (Grimsdell 1969). For zebra, a modification of Klingel and Klingel's method (1966) was used (Spinage 1972). Warthog determinations up to 4 years of age were based on information supplied by G. Clough (personal communication), and the data of Child, Sowls, and Mitchell (1965). Above this age, determinations were made according to Spinage and Jolly (1972).

RESULTS

The collections included 458 impala skulls, 451 warthog, 144 zebra, 133 buffalo, 62 topi *Damaliscus korrigum* Ogilby, 54 eland *Taurotragus oryx* Pallas, and 47 waterbuck *Kobus defassa ugandae* Neumann. The ratios of female to male skulls are shown in Table 1, and age distributions in Fig. 1. Of those skulls which could be sexed with certainty, the ratio approached 1:1 in the larger species. This suggested departures from this ratio were partly a reflection of differences in skull destructibility rather than disparate sex ratios. In hornless females, hyaenas *Crocuta crocuta* Erxleben can carry the skull away with ease and chew up the cranium, leaving little behind (Sutcliffe 1970). In the Uganda waterbuck,

TABLE 1. Sex ratio of examined skulls

Species	Numbers of skulls	Ratio female: male
Impala	107:334	1:3
Topi	22:40	1:1.8[a]
Waterbuck	18:29	1:1.6[a]
Eland	19:25	1:1.3[a]
Buffalo	52:68	1:1.3
Zebra	45:51	1:1.1

[a]Numbers too small for life table analysis.

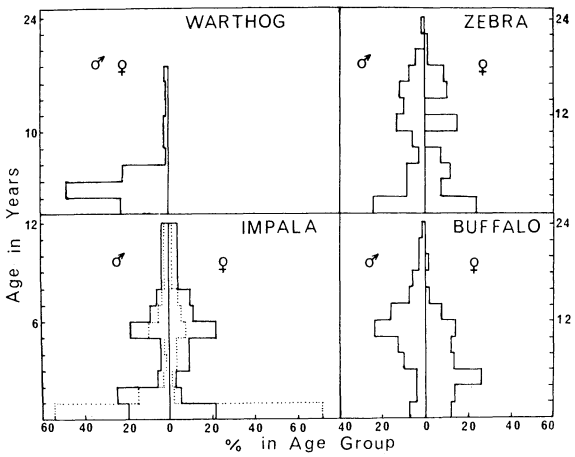

FIG. 1. Age-distribution pyramids of collected skulls of four main species. Male impala show "correction factor" (2) applied, dotted line (see Table 4). Female impala show 50% mortality in first age group, dotted line. All other distributions are raw data, but equal cohorts of sexes are shown.

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TABLE 2. Proportion of juvenile to adult skulls of four main species

Species	Juvenile stage in years[a]	Number of juvenile skulls	Number as % of total skulls
Impala	0–2	40	17%
Buffalo	0–3	19	17%
Warthog	0–2	99	22%
Zebra	0–3	38	41%

[a]A subjective category based on lightness of the skull bones and stage of suture fusion.

TABLE 3. Observed numbers of male impala in a live population, compared with examined skulls

	Estimated age				
	3 mo–1 yr	1–2 yr	2–3 yr	> 3 yr	Total[a]
Number counted	116	80	123	300	619
Skulls	18	22	5	289	334

[a]Ratio of numbers < 3 years to > 3 years: 1:0.94 among living animals, versus 1:6.42 among skulls.

a species with marked sexual dimorphism of the skull, female skulls were under represented in a collection by almost 1:3 (33:104), compared to an estimated live sex ratio of 1.6:1 (Spinage 1970).

Discrepancy also occurs between the number of juvenile skulls collected and those of adults, as the young skulls are more easily destroyed. An exception was the zebra, in which even foal skulls are fairly large and durable (Table 2).

Impala

In 1958 nearly 100 ha of lakeside area were made available to the animal populations. This habitat is particularly favored by impala, and their numbers appear to have increased four-fold as a result (Spinage and Guinness 1972). Experience shows, given favorable conditions, the initial rate of increase of an expanding population in a competitive environment is rapid, followed by a decreasing rate of recruitment until mortality and recruitment approximate a stationary population. It is not possible to demonstrate that this has occurred in this population without further population counts, but the species has had sufficient time to respond to the changed circumstances so that its population may have become stationary.

As this antelope is relatively small, young age classes are likely to be deficient in skull collections. Calves are susceptible to a wider range of predatory mammals, as well as to predatory birds. For males, a correction factor was applied by reference to a count of live animals, as from 3 months to 3 years males can be roughly separated into age groups according to horn shape and size. Two counts were made, in March and April, when the previous season's July peak of births would have been 9 months old. In contrast to the skull collection, which had 6.42 times as many skulls greater than 3 years old as younger, there were 1.06 times as many animals less than 3 years old as there were animals above this age (Table 3). If the population is stationary, the number of animals dead is inversely related to the number living; to correct the discrepancy, the number of skulls in each group less than 3 years of age was multiplied by 6.8 (6.42×1.06). This is correction factor (1) in Table 4.

Animals less than 3 months of age were not included in the live count as it was not possible to distinguish the sexes readily; but many ungulates show at least 50% mortality in the first year of life, so that a further correction factor of this order was applied. This is correction factor (2) in Table 4.

Precise inferences cannot be drawn from such data, because the first "correction factor" confuses a mixture of d_x and l_x data, and the second factor is arbitrary. But in actuality such factors would have a considerable effect upon the expectation of life at birth. Less marked would be the effect upon the mean expectation of life, for the factors have their effect merely where they are applied. Adult mortality is unaltered, except that the size of the hypothetical cohort is lowered accordingly, accentuating the initial rate of decline in numbers.

Thirty per cent fewer female than male skulls were recovered less than 3 years old, and 70% less in the remaining age classes. This initial 30% difference results in correction factors (1) and (2) depressing the female curve to approximate closely the male with factor (2) applied. An initial 50% mortality applied to the crude female data produces a curve approximating the male curve with factor (1) applied. As these adjusted female curves follow the male curves closely, they are not shown in Fig. 2.

Buffalo

Studies in Uganda suggested that a buffalo population could stabilize its numbers in about 9 years, consequent upon a favorable habitat change (Laws 1968*a*). It is likely that the population in the Akagera Park would also have stabilized in numbers by 1968, had there been an increase since the habitat change in 1958.

Several studies (Pienaar 1969, A. Sinclair, personal communication), suggest a 50% mortality in the first year of life to be a reasonable assumption. Numbers in the first-year class were thus corrected by this factor (Table 5). Fewer female than male skulls were recovered, but there were more young ones, indicating a shorter expectation of life. From 1 to 10 years the ratio of males to females was 27:57, and

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TABLE 4. Life tables of the impala, 1967–69

x	d_x	kl_x	kq_x	kL_x	kT_x	e_x[g]
			Males[a]			
0–1	18	1000	54	1271[f]	6018	6.0
1–2	22	945	70	912	4746	5.0
2–3	5	880	21	872	3834	4.3
3–4	22[e]	865	76	832	2962	3.4
4–5	22[e]	800	82	768	2129	2.6
5–6	109	735	225	575	1316	1.8
6–7	53	415	255	332	786	1.9
7–8	39	249	470	190	454	1.8
8–9	11[e]	132	250	115	264	2.0
9–10	11[e]	99	330	82	148	1.5
10–11	11[e]	66	500	50	66	1.0
11–12	11[e]	33	1000	16	16	0.5
Mean length of life = 6.2 years					$\bar{e}_x$ = 2.65	
			Males[b]			
0–1	122	1000	205	1167[f]	4008	4.0
1–2	149	795	314	669	2842	3.6
2–3	34	543	105	514	2172	4.0
3–4	22[e]	485	76	467	1658	3.4
4–5	22[e]	450	82	431	1191	2.6
5–6	109	412	445	320	760	1.8
6–7	53	228	490	184	440	1.9
7–8	39	140	468	107	256	2.0
8–9	11[e]	74	250	65	149	2.0
9–10	11[e]	56	333	46	84	1.5
10–11	11[e]	37	500	28	38	1.0
11–12	11[e]	19	1000	10	10	0.5
Mean length of life = 4.25 years					$\bar{e}_x$ = 2.36	
			Males[c]			
0–1	472	1000	500	975[f]	2764	2.8
1–2	149	500	314	421	1789	3.6
2–3	34	342	105	324	1368	4.0
3–4	22[e]	305	76	294	1044	3.4
4–5	22[e]	283	82	272	750	2.7
5–6	109	260	445	202	479	1.8
6–7	53	144	490	116	277	1.9
7–8	39	88	468	68	161	1.8
8–9	11[e]	47	250	41	94	2.0
9–10	11[e]	35	333	29	52	1.5
10–11	11[e]	23	500	18	29	1.0
11–12	11[e]	12	1000	6	6	0.5
Mean length of life = 3.0 years					$\bar{e}_x$ = 2.25	
			Females[d]			
0–1	23	1000	199	1179[f]	5195	5.2
1–2	6	805	64	779	4022	4.7
2–3	3	753	34	740	3242	4.3
3–4	9[e]	728	108	689	2502	3.4
4–5	9[e]	650	120	610	1813	2.8
5–6	23	570	348	471	1203	2.1
6–7	16	372	371	303	732	2.0
7–8	12	234	445	182	429	1.8
8–9	4[e]	130	266	112	247	1.9
9–10	4[e]	95	363	78	134	1.4
10–11	4[e]	61	570	44	56	0.9
11–12	3[e]	26	1000	13	13	0.5
Mean length of life = 5.4 years					$\bar{e}_x$ = 2.58	

[a]Raw data: no correction factors applied.
[b]Correction factor (1) applied: 6.8 times classes 0–3 years. See text for justification.
[c]Correction factor (1) applied, also correction factor (2): 50% mortality assumed in first year of life.
[d]Tabled values are raw data. "Correction factor" (1) applied to the female data gives an expectation of life at birth of 2.6 years; mean length of life of 2.65 years, and e_x = 2.22 years. An initial 50% mortality in class 0–1 applied to the crude data, gives an expectation of life at birth of 2.9 years; a mean length of life of 3 years, and $\bar{e}_x$ = 2.42 years.

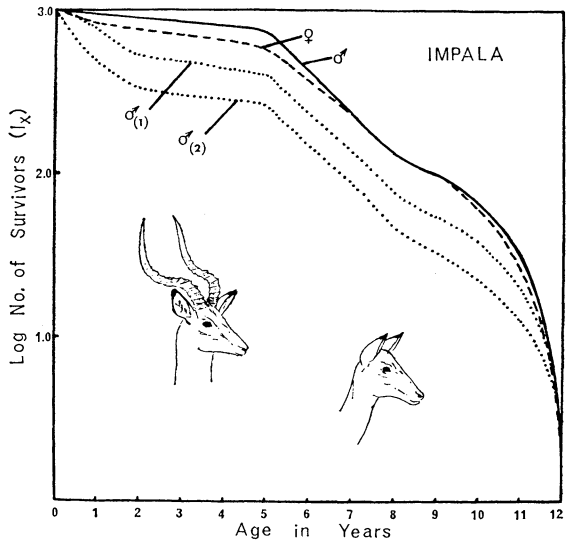

FIG. 2. Survival curves of impala, smoothed data. Continuous line, males; broken line, females. Dotted line shows "correction factors" (1) and (2) applied to the male curve (see Table 4).

from 11 to 17 years this was completely reversed to 39:16. This difference in survival is probably real, as we would expect to find fewer young female skulls than male skulls, due to their greater destructibility. Grimsdell (1969) concluded there was no difference in age determination between the sexes, so such discrepancy is unlikely to be due to this. The alternative suggests a strong survival factor operating against the female, centered around the 4th to 7th years of life. Males have a better expectation of life at all ages (Table 5 and Fig. 3).

Zebra

Estimates of 1,400 zebra in 1947 (Verhulst, quoted by Akeley 1950), and of 1,500 in 1968 by myself, suggest the population may be stationary. But data are lacking as to whether numbers have fluctuated widely in the interim.

Skulls of young animals are large, and almost as easily found as those of adults. No correction factor has thus been applied to the yearling class, although there is almost certainly some underrepresentation. The pattern of survival was similar for both sexes (Table 6 and Fig. 4). The population is subject to hunting in an adjacent reserve, and a peak in numbers of 234 animals was shot in 1965–66. This does not appear to be reflected in the survival data.

[e]In the classes indicated, a lumped age determination has been divided between two or more classes.
[f]A separation factor of 0.3 has been used in calculating kLx at age 0–1 years.
[g]Standard demographic notation is used. x = age in years; d = number of deaths; l_x = the number surviving from one age class to the next; q_x = the rate of mortality; L = the number of animals alive between age x and $x + 1$; T = the total number of animls x age units beyond age x, which is used to calculate e, the expectation of life ($e_x = T_x/l_x$). The value of k is a constant determining the size of the hypothetical cohort, here 1,000.

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TABLE 5. Life tables of the buffalo, 1967–69[a]

x	d_x	kl_x	kq_x	kL_x	kT_x	e_x
			Males			
0–1	2 (70)[b]	1000	500	975	6025	6.0
1–2	3	500	43	484	5050	10.0
2–3	3	469	45	464	4566	9.9
3–4	0	458	0	458	4102	9.0
4–5	0	458	0	458	3644	8.0
5–6	3	458	47	446	3186	7.0
6–7	1	435	11	432	2740	6.3
7–8	6	430	100	408	2307	5.4
8–9	3	386	55	375	1899	4.9
9–10	6	364	118	342	1524	4.2
10–11	8	321	178	292	1182	3.7
11–12	10	264	270	228	889	3.4
12–13	7	193	260	168	660	3.4
13–14	5	143	250	125	492	3.4
14–15	2	107	134	100	368	2.6
15–16	3	93	230	82	268	2.9
16–17	4	71	400	57	185	2.6
17–18	1[c]	43	167	40	128	3.0
18–19	1[c]	36	200	32	89	2.5
19–20	1[c]	29	250	25	56	2.0
20–21	1[c]	21	330	18	32	1.5
21–22	1[c]	14	500	10	14	1.0
22–23	1[c]	7	1000	4	4	0.5
Mean length of life = 6.3 years						$\bar{e}_x$ = 4.48
			Females			
0–1	3 (55)[b]	1000	500	975	4011	4.0
1–2	4	500	72	482	3036	6.0
2–3	4	465	78	446	2553	5.5
3–4	4	428	85	410	2107	5.0
4–5	4	392	93	374	1697	4.3
5–6	11	355	285	305	1323	3.7
6–7	4	255	144	237	1018	4.0
7–8	4	219	166	200	781	3.6
8–9	4	182	200	164	581	3.2
9–10	3	145	200	131	417	2.9
10–11	3	118	250	104	286	2.4
11–12	5	91	460	68	181	2.0
12–13	2	46	427	36	113	2.5
13–14	1	27	500	22	76	2.8
14–15	0	18	0	18	54	3.0
15–16	1	18	500	14	36	2.0
16–17	0	9	0	9	22	2.5
17–18	0	9	0	9	13	1.5
18+	1	9	1000	4	4	0.5
Mean length of life = 4.27 years						$\bar{e}_x$ = 3.23

[a]Notation as for table 4.
[b]Figure in parentheses assumes 50% mortality in age class 0–1.
[c]Indicates lumped age determination divided among two or more age classes.

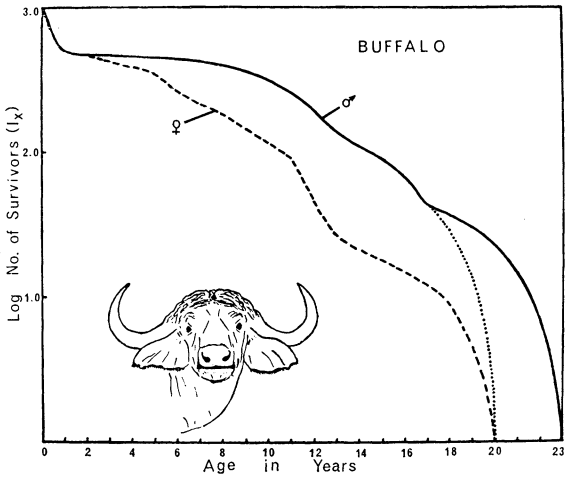

FIG. 3. Survival curves of buffalo, smoothed data. Continuous line, males; broken line, females. Dotted line shows course of male survival when the oldest age classes are lumped.

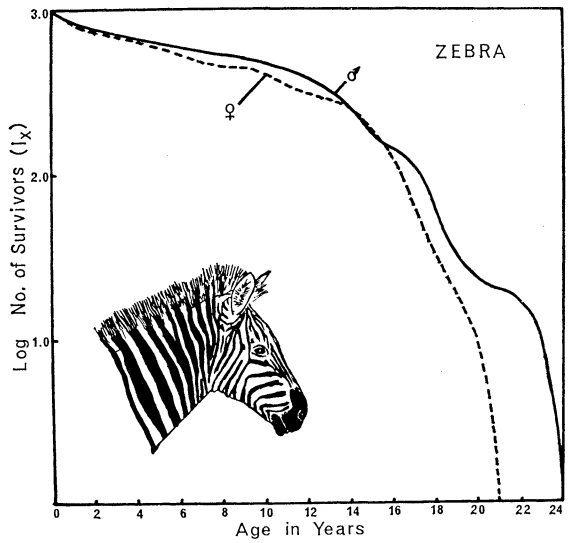

FIG. 4. Survival curves of zebra, smoothed data. Continuous line, males; broken line, females.

Warthog

This species has probably increased in numbers since 1958, inasmuch as the lake regions, which became available then, are its favored habitat. Its burrowing habit precluded any precise estimation of its numbers, either then or now. However, the reproductive rate is high so that a stationary population might be expected to result early, consequent upon any opportunity for increase occurring.

In Uganda, Clough (personal communication) could detect no criteria of sexual dimorphism in warthog skulls. Since reliable differentiation was not possible, a composite life table for both sexes was compiled (Table 7 and Fig. 5). Clough (personal communication) estimated there was 50% mortality in the first year of life in the population he studied, and the same factor has been applied here.

Although the potential duration of life in the warthog is long, the species seems to undergo a fairly rapid life cycle with a high reproductive rate. The smoothed survival curve showed a marked exception to that usually associated with large mammals. There was almost random mortality to middle-age, when the few remaining survivors fitted a negatively skewed pattern. For the younger ages this was similar to Clough's preliminary analysis of his Uganda population (personal communication).

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TABLE 6. Life tables of zebra, 1967–69[a]

x	d_x	kl_x	kq_x	kL_x	kT_x	e_x
			Males			
0–1	12	1000	179	1184	9038	9.0
1–2	4	822	73	791	7854	9.5
2–3	2	760	39	745	7063	9.2
3–4	4	730	81	700	6318	8.7
4–5	2	671	44	656	5618	8.4
5–6	4	641	93	611	4962	7.7
6–7	1	581	25	575	4350	7.5
7–8	1	569	26	560	3776	6.6
8–9	3	551	81	530	3216	5.8
9–10	1	508	29	500	2686	5.7
10–11	4	492	121	462	2186	4.5
11–12	5	433	172	396	1724	4.0
12–13	2	358	84	343	1328	3.7
13–14	5	328	226	291	985	3.0
14–15	6	254	354	209	694	2.7
15–16	2	164	182	149	485	3.0
16–17	1	134	111	126	336	2.5
17–18	4	119	500	90	210	1.8
18–19	2	60	500	45	120	2.0
19–20	1	30	500	22	75	2.5
20–21	0	15	0	15	52	3.5
21–22	0	15	0	15	38	2.5
22–23	0	15	0	15	22	1.5
23–24	1	15	1000	8	8	0.5
Mean length of life = 9.72 years					$\bar{e}_x$ = 4.8	
			Females			
0–1	12	1000	196	1173	8235	8.2
1–2	4	805	82	772	7062	8.7
2–3	4	738	89	705	6290	8.5
3–4	1	672	24	664	5585	8.3
4–5	2	656	50	640	4921	7.5
5–6	6	625	168	575	4280	6.8
6–7	2	525	62	509	3706	7.1
7–8	3	493	100	468	3196	6.5
8–9	0	443	0	443	2728	6.1
9–10	0	433	0	443	2286	5.1
10–11	6	443	223	394	1842	4.2
11–12	4	345	190	312	1448	4.2
12–13	0	278	0	278	1137	4.1
13–14	0	278	0	278	859	3.1
14–15	6	278	353	229	581	2.1
15–16	3	180	274	156	352	2.0
16–17	4	131	500	98	196	1.5
17–18	2	66	500	50	98	1.5
18–19	1	33	500	24	48	1.5
19–20	0	16	0	16	24	1.5
20–21	1	16	1000	8	8	0.5
Mean length of life = 8.46 years					$\bar{e}_x$ = 4.7	

[a]Notation as for Table 4.

TABLE 7. Life table of the warthog, sexes combined, 1967–69[a]

x	d_x	kl_x	kq_x	kL_x	kT_x	e_x
0–1	10 (423)[b]	1000	500	975	2341	2.3
1–2	89	500	210	447	1366	2.7
2–3	105	394	315	332	919	2.3
3–4	104	270	454	209	587	2.2
4–5	78	148	625	102	378	2.6
5–6	11	56	234	50	276	4.9
6–7	5	43	139	40	226	5.3
7–8	1	37	32	36	186	5.0
8–9	4	35	133	33	150	4.3
9–10	4	31	154	28	118	3.8
10–11	3	26	136	24	89	3.4
11–12	5	22	263	16	65	3.0
12–13	3	16	215	14	49	3.1
13–14	1	13	91	12	34	2.7
14–15	2	11	200	10	22	2.0
15–16	1	9	125	8	12	1.4
16–17	7	8	1000	4	4	0.5
Mean length of life = 2.62 years					$\bar{e}_x$ = 3.03	

[a]Notation as for Table 4.
[b]Figure in parentheses assumes 50% mortality in age class 0–1.

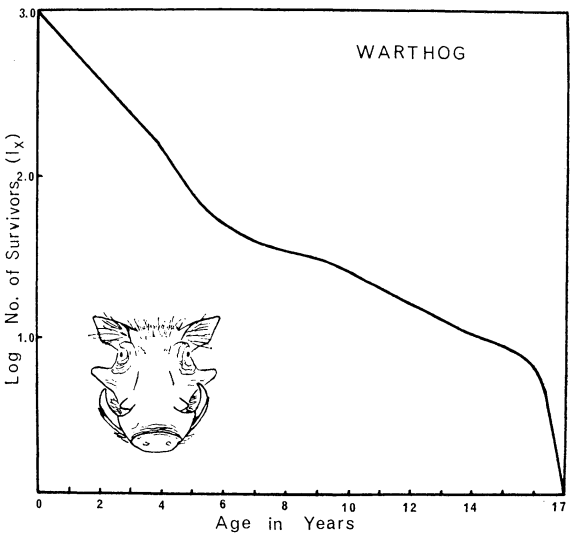

FIG. 5. Survival curve of warthog, smoothed data. Sexes combined.

DISCUSSION

A major drawback to natural survival data is the underrepresentation of young age classes, commented upon by several authors (Bourliére 1959, Caughley 1966). This is also the major source of error in human life tables, due to underregistration of infant deaths (Barclay 1958). In wildlife studies the deficiency can be countered by direct methods of counting yearlings and sampling for fertility. One approach to minimizing this source of error in continuous breeders was suggested by Spinage (1968). For a seasonal breeder in which calving is spread over several months, Sinclair (1969) used an adaptation of Richards and Waloff's method (1954). But this method is inapplicable to continuous breeders and to species which grow rapidly in early life so that stages of growth cannot be readily separated.

Unless based on an actual cohort of animals, all born at the same time and followed to the death of the last survivor, a life table represents a synthetic cohort. What is presented as a single cohort is a composite of as many cohorts as there are age classes in the life table, each having fixed rates of mortality. The life table is therefore time specific. A synthetic cohort is not the hypothetical cohort, the arbitrary

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radix used for analysis, usually 1,000. From a real cohort an age-specific life table can be constructed, the cohort being followed from stage to stage. With found skulls the procedure is reversed; deaths are added from the last survivor backwards, to finish with an assumed initial size of cohort.

Populations contain relatively few old animals, so that such backward projections incorporate little error, and imperfections in age determination at this end of the scale will have little effect. But under-representation of young classes means that a single meaningful figure, with which to compare different populations, cannot be calculated. As the stages in a time-specific life table are unaffected by the antecedent stages, a figure can be calculated starting at any intermediate stage, but this will tell us little about the population as a whole.

Precision in chronological age separation is less important than the numbers within each category, so that the shape of the survival curve is unlikely to be seriously affected by relatively empirically crude methods of age determination. In this respect the curve is robust, but this limits its usefulness when accuracy is required.

The extent to which the size of the skull sample is representative of a population depends upon the size of the population, and its life span. In a natural population, exposed to scavengers, it is unlikely that more than 5% of the population would be recovered at any one time. From an estimated population of 3,000–4,000 elephants *Loxodonta africana* Blumenbach, Laws (1966) recovered only 133 skulls, representing 3%–4% of the total population; and elephant skulls are not easily overlooked.

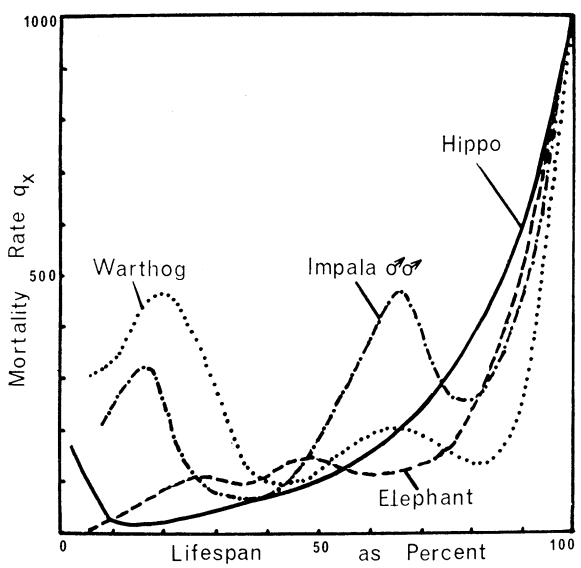

FIG. 6. Mortality rate (q_x). Origins for all species would probably commence at about the 500 level. Data grossly smoothed. Elephant and hippopotamus data from Laws (1966, 1968).

Caughley (1966) suggested that many of the shortcomings of the survival curve could be overcome by using the q_x (rate of mortality) curve, as the points are independent of all but two age frequencies. However, the q_x curve requires adequate sampling of all age classes; otherwise a very erratic curve results. In Fig. 6, grossly smoothed q_x curves of four species are shown. It is difficult to know whether the irregularities represent sampling deficiencies or express the true rates of mortality.

Restrictions of the life table data

Unless certain restrictive assumptions are met, a time-specific life table can represent no more than a crude generalization of the probable population structure, at the time the data were collected. It cannot tell us how the structure was attained, nor what its future pattern might be. As time-specific life tables represent a stationary population, the net reproductive rate (R_o) is unity, and the intrinsic rate of natural increase (r_m) is zero. If the population is increasing or decreasing, this is likely to be reflected in the survival curve. But as the data are forced into a model in which the number of deaths is equal to the number of births ($l_o = \Sigma d_x$), the distribution is distorted, and the survival curve is no longer a correct representation of the age distribution of the population.

It is impossible to tell from a single life table whether or not the population is stationary, but this can be determined from other data, such as repeated censuses. A stationary population is one that does not change in size because mortality equals natality, but its age distribution could adopt any pattern, at least temporarily. This is unlike the stable age distribution, which is the age structure of a population that has achieved a constant rate of change. Thus the stable age distribution has no relevance to time-specific life tables.

The inferences that may be drawn from life tables derived from skull collections are therefore limited. Provided these limitations are borne in mind, and the data regarded as a crude method of identifying gross differences in population age structure at a given time, as distinct from trends within populations, they still provide a valid analysis.

Comparison of the survival data

The survival curves of the four species examined in the Akagera Park are compared in Fig. 7 by plotting number of survivors against lifespan as a percentage. Caughley (1966) has discussed the disadvantages of using percentage deviation from mean length of life for making such comparisons. The impala, buffalo, and zebra show the general form of adult survival curve described by Pearl (1940) as a negatively skewed rectangular curve. The warthog data suggest wide departure from this typical curve

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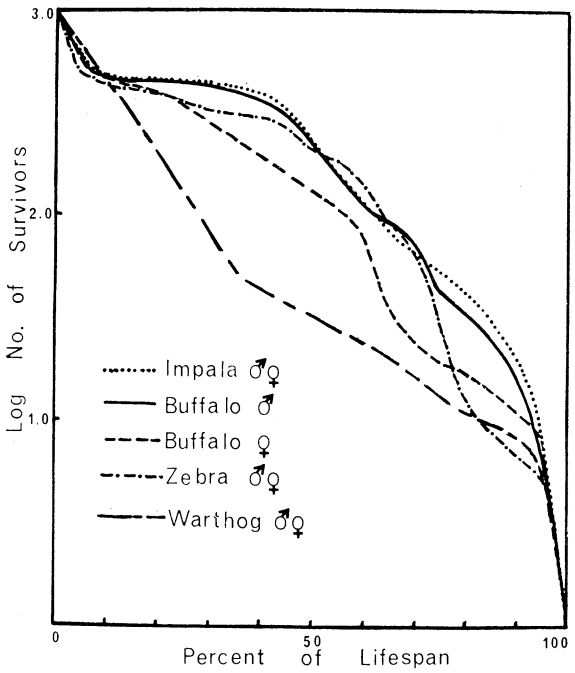

FIG. 7. Survival curves of four main species compared, smoothed data. All curves are taken from an initial 50% mortality.

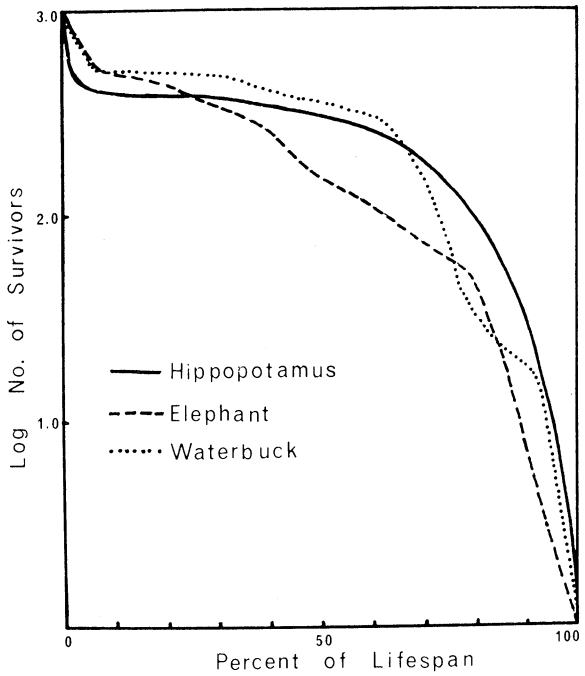

FIG. 8. Survival curves of three other populations compared, smoothed data. All curves are taken from an initial 50% mortality. Elephant and hippopotamus data from Laws (1966, 1968) for populations in the Murchison Falls National Park and Queen Elizabeth National Park, Uganda, respectively. Waterbuck data from Spinage (1970), Queen Elizabeth National Park. Sexes are combined for all species.

is possible, the warthog apparently showing an age-constant death rate for much of the lifespan, usually characteristic of small, short-lived, fast-breeding species. Child, Roth, and Kerr (1968) have recorded high initial mortality in warthog populations in Rhodesia, suggesting this type of pattern.

The close similarity of the impala and male buffalo curves suggests that small size—and so relative ineffectualness of defense against large predators, as well as exposure to smaller and hence a greater range of predators—is not an important factor in shaping the pattern of survival. This has also been observed by Caughley (1966).

Three species, from published sources, are compared in Fig. 8. Two of these are long lived; the hippopotamus *Hippopotamus amphibius* L. reaches about 45 years (Laws 1968), and the elephant reaches about 64 years (Laws 1966). Also shown is a waterbuck population, with a maximum age of about 12 years (Spinage 1970). This comparison confirms the typical form of the survival curve in undisturbed populations of large ungulates and shows that it is not related to length of lifespan.

The warthog differs from the other species considered, not in its potential length of life, which is 16–17 years (Spinage and Jolly 1972), but in its high reproductive rate. The other species are all monovular, with birth intervals ranging from 10 months to 4 years; but the warthog is polyovular, with annual litters of 2–7 offspring (Clough 1969). This high reproductive rate is offset by a high mortality rate, an animal having an almost equal chance of dying at any age up to middle-age. In contrast, the chances of survival of the other ungulates tend to increase over this period. It is the reproductive rate which thus apparently determines the pattern of survival, not bodily size or length of lifespan.

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